

Spotify Customer Churn Data Analysis

About Dataset:

- **Purpose:** To predict whether a Spotify user will churn (cancel subscription) or remain active.
- **Rows:** Each row represents a unique Spotify user
- **Columns (Features)**
- **user_id** → Unique identifier for each user
- **gender** → User gender (Male/Female/Other)
- **age** → User age
- **country** → User location
- **subscription_type** → Type of Spotify subscription (Free, Premium, Family, Student)
- **listening_time** → Minutes spent listening per day
- **songs_played_per_day** → Number of songs played daily
- **skip_rate** → Percentage of songs skipped
- **device_type** → Device used (Mobile, Desktop, Web)
- **ads_listened_per_week** → Number of ads heard per week
- **offline_listening** → Offline listening minutes
- **is_churned** → Target variable (0 = Active, 1 = Churned)
- **Dataset Type:** Mixed (numeric + categorical)

Use Case: Build machine learning models to predict user churn, analyze engagement patterns, and help Spotify reduce cancellations.

Spotify Customer Churn Analysis Report:

This project focuses on analyzing **Spotify user data** to predict churn and provide actionable insights to improve user retention.

Key Insights:

- ❖ **Churn Rate** – ~25.9% of users churned, while 74.1% remained active. This indicates **1 in 4 users cancel their subscription**, showing a moderate churn risk.

- ❖ **Subscription Type Impact** – Family users show the **highest churn risk (~27.5%)**, while Free (~24.9%), Premium (~25.1%), and Student (~26.2%) are almost similar, meaning subscription type impacts churn but not drastically.
 - ❖ **Age Group Distribution** – Adults (26–50 yrs) are the **largest user segment (56.2%)**, followed by Young (22.9%) and Senior Citizens (20.8%). This shows Spotify's **core audience is adults**.
 - ❖ **Gender Distribution** – Almost evenly distributed among Male (33.6%), Female (33.2%), and Other (33.1%), suggesting churn is **not strongly influenced by gender**.
 - ❖ **Device Preference** – Desktop users dominate slightly (34.7%), while Web (32.8%) and Mobile (32.5%) are close, indicating **multi-device engagement is common**.
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Final Conclusion

To **reduce churn and increase engagement**, Spotify should focus on:

- ❖ The churn rate is significant (1 in 4 users), and **subscription type has more influence than demographics or device usage**.
 - ❖ Since listening time doesn't differ much between active and churned users, **content quality, engagement features, or pricing strategy** might be bigger churn drivers than pure usage.
 - ❖ Retention strategies should focus on **subscription-specific engagement**, particularly **Free and Family plans**, to reduce churn while keeping Premium users loyal.
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Project Information

- **Project Name:** Spotify Customer Churn Analysis
 - **Tools Used:** Python (Pandas, NumPy, Matplotlib)
 - **Report Type:** User Engagement & Churn Prediction
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This analysis helps Spotify identify high-risk users, improve retention strategies, and make data-driven decisions to reduce cancellations.